1. Executive Summary & Big Picture Overview

The dataset under review contains 327 raw transaction records collected from a fintech platform during the calendar year 2025, covering the period from the sixth of January to the thirtyfirst of December. After removing five exact duplicate rows, the clean set consists of 322 unique transactions, of which 314 have valid monetary amounts. All amounts are expressed in Indian Rupees, with a total portfolio value of three million eight hundred twentysix thousand six hundred seventynine rupees and fortythree paise.

Monthly volume and revenue show a clear seasonal pattern. The highest monthly revenue appears in May 2025, with a total of three hundred and four thousand three hundred seventyfive rupees and a mean transaction value of eighteen thousand eight hundred and thirtysix rupees. The lowest revenue month is September 2025, with one hundred and fiftynine thousand one hundred and one rupees and a mean of six thousand three hundred sixtyfour rupees. The overall mean transaction amount across the year is twelve thousand one hundred eightysix rupees and eightyeight paise, while the median - the middle value when all amounts are sorted - is four thousand seven hundred eightytwo rupees and eightynine paise. The smallest transaction recorded is one hundred fiftyfour rupees and ninetynine paise, and the largest is four hundred fifty thousand rupees.

These figures suggest a healthy portfolio with a broad range of transaction sizes. The presence of a few very large transactions (up to four hundred fifty thousand rupees) is balanced by a large number of smaller, routine payments, indicating diversified usage of the platform across different customer segments.

2. Data Quality & Missing Information Audit

Data integrity is critical for reliable analytics. The audit reveals five duplicate rows, all sharing the same transaction identifiers: TXN202500021, TXN202500076, TXN202500102, TXN202500151, and TXN202500251. Removing these duplicates leaves 322 distinct records.

Missing values are distributed unevenly across columns. The reference number column has four missing entries, transaction amount has four, category has five, status has four, and account ID has five. No missing values appear in transaction ID, transaction date, currency, transaction type, channel, or customer region. In total, 22 cells are empty.

Additionally, five dates fail to conform to the expected format, and four amounts contain invalid numeric values. These anomalies can distort statistical summaries and hinder downstream processing. Clean records are essential because they ensure that calculations such as averages, medians, and outlier detection are based on accurate data, thereby preventing misleading conclusions and reducing the risk of erroneous business decisions.

3. Transaction Statistics & Granular Breakdown

The dataset's core statistics are as follows: 314 valid transactions, a total of three million eight hundred twentysix thousand six hundred seventynine rupees and fortythree paise, a mean of twelve thousand one hundred eightysix rupees and eightyeight paise, a median of four thousand seven hundred eightytwo rupees and eightynine paise, a minimum of one hundred fiftyfour rupees and ninetynine paise, and a maximum of four hundred fifty thousand rupees.

Percentiles provide a deeper view of the distribution. The first quartile (p25) is two thousand twentythree rupees and twenty paise, the median (p50) is four thousand seven hundred eightytwo rupees and eightynine paise, and the third quartile (p75) is nine thousand five hundred eightynine rupees. The interquartile range (IQR), which measures the spread of the middle fifty percent of values, is seven thousand four hundred sixtyfive rupees and eighty paise.

Monthly trends show that transaction volume peaks in July 2025 with 28 transactions and a total of six hundred thirty thousand one hundred six rupees, while the lowest volume occurs in September 2025 with 25 transactions and a total of one hundred fiftynine thousand one hundred one rupees.

By category, Merchant Purchase dominates with 112 transactions and a total of one million three hundred fiftysix thousand four hundred fiftyeight rupees, followed by Fund Transfer with 77 transactions and a total of one million seven hundred ninetythree thousand four hundred fortyseven rupees. Bill Payment, Cash Withdrawal, Missing, Refund, and Subscription categories account for the remaining transactions, each with distinct average amounts and medians.

Status distribution shows that Completed transactions form the majority (272 records) with a total of three million five hundred fortyseven thousand five hundred fiftyfour rupees. Pending, Failed, Reversed, and Missing statuses account for the remainder, each with their own average and median values.

Channel usage indicates that Mobile is the most common channel (107 transactions, total of one million six hundred sixtyfive thousand five hundred fortyfour rupees), followed by Web, API, ATM, POS, and Auto Debit, each with varying transaction counts and average amounts.

Regional activity reveals that the North region processes the highest total amount (one million fiftythree thousand nine hundred thirtythree rupees), followed by Central, West, East, and South, each with distinct transaction counts and average values.

4. Unusual Activity & Giant Spikes

Outlier detection was performed using the 1.5 times IQR rule, a simple method that flags any transaction whose amount lies more than one and a half times the IQR above the third quartile or below the first quartile. In plain terms, if a transaction is significantly larger or smaller than the typical middle range, it is flagged for review.

The lower fence is negative nine thousand one hundred seventyfive rupees and fifty paise, effectively meaning no transaction can be below zero, so only the upper fence matters. The upper fence is twentysix thousand eight hundred and seven rupees and seventy paise. Any transaction exceeding this amount is considered a review candidate.

Twentyseven transactions meet this criterion. The most extreme case is a Merchant Purchase on July 13, 2025, for four hundred fifty thousand rupees, which far exceeds the upper fence. Other highvalue transactions include Fund Transfers of three hundred thousand rupees, two hundred and seventyfive thousand rupees, and two hundred and thirtyfive thousand rupees, all occurring between March and November 2025. Several transactions around twentynine thousand rupees also exceed the upper fence, indicating that the rule captures both very large and moderately large amounts that may warrant closer inspection.

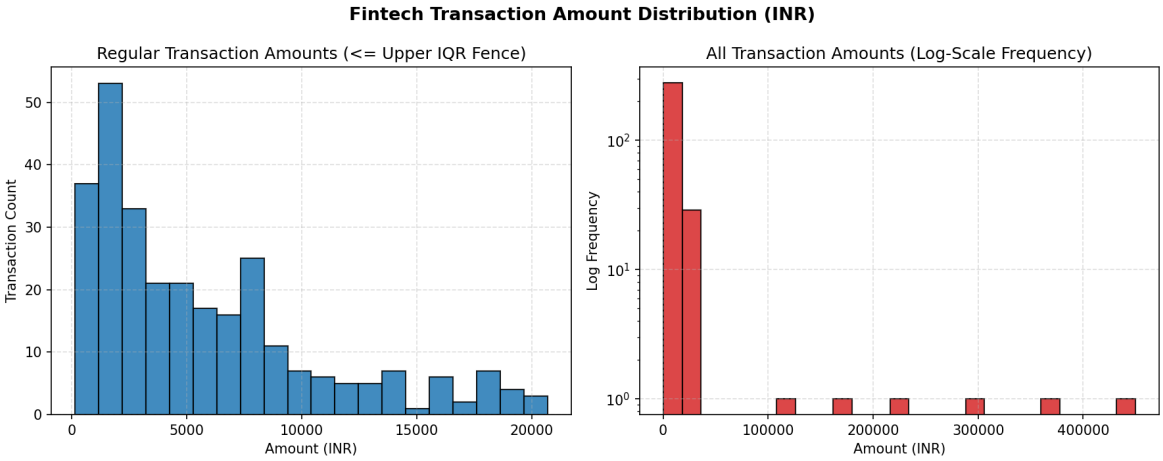
Each flagged transaction includes details such as transaction ID, date, amount, category, status, channel, and customer region, providing context for analysts to determine whether the transaction is legitimate or potentially fraudulent.

5. Actionable Advice for Risk Mitigation & Data Pipelines

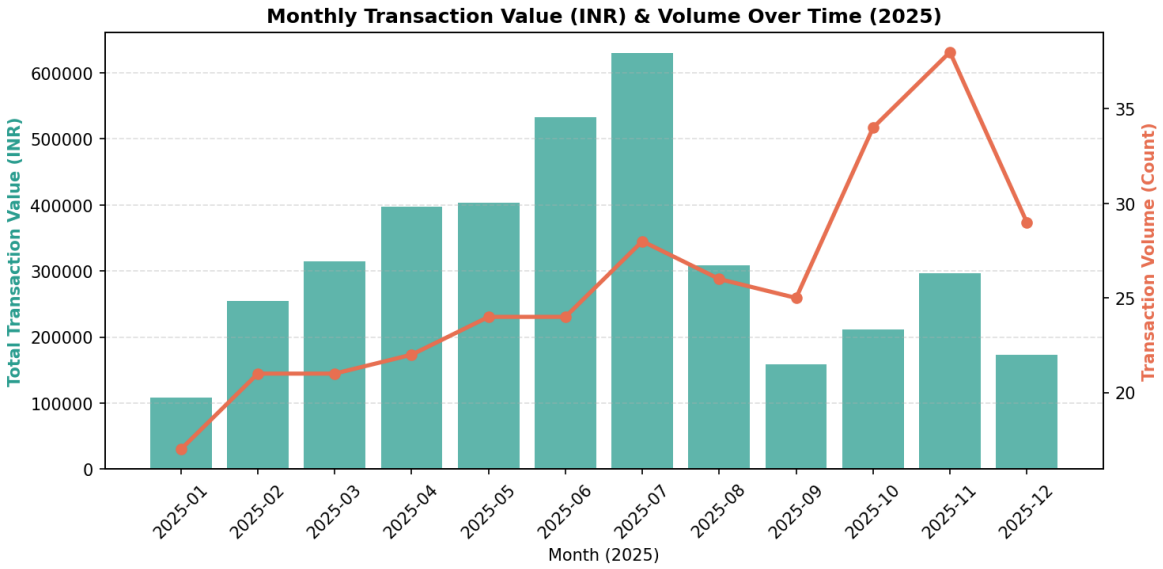
- Duplicate Removal: Implement an automated deduplication step that checks transaction identifiers before ingestion to prevent duplicate records from inflating counts and totals.
- Missing Value Handling: Enforce mandatory fields for reference number, transaction amount, category, status, and account ID. For any missing entries, trigger a data quality alert and request manual correction or automated imputation only when business rules allow.
- Date and Amount Validation: Add validation logic that rejects or flags dates that do not match the YYYYMMDD format and amounts that are negative or exceed realistic bounds.
- Outlier Monitoring: Deploy a realtime monitoring dashboard that flags transactions above the upper IQR fence. Include contextual information such as channel and region to aid rapid triage

Visual Analytics & Chart Artifacts

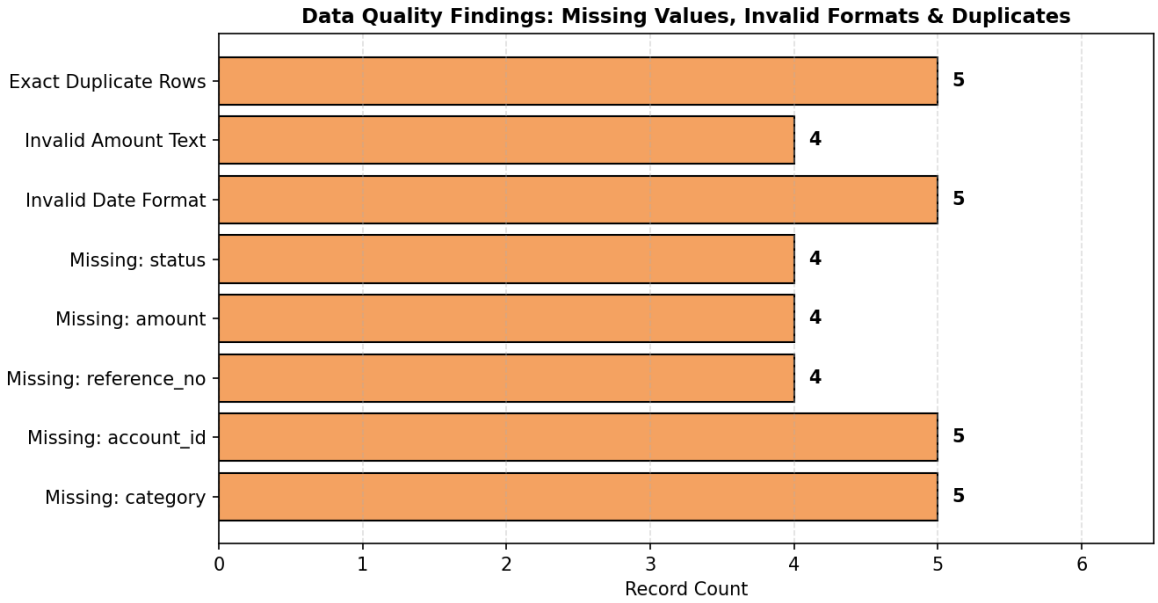
Amount Distribution (IQR & Spikes)



Transaction Volume Over Time



Missing Values Audit



Category & Channel Breakdown

